Bioinformatics, 2026, 1–16
doi:

Supplementary Material for: TissuePMHC: Unveiling Tissue-Specific Preferences in MHC-I Presentation

Yuheng Cai,[1,2,3,†] Liuxintong Zhang,[1,†] Lance Turtle[2] and Jia Meng[1,*]

[1]Department of Biosciences and Bioinformatics, School of Science, Xi'an Jiaotong-Liverpool University, Suzhou, China, [2]Institute of Infection, Veterinary and Ecological Sciences, Liverpool, UK and [3]Themedium Therapeutics Co. Ltd., Suzhou, China

[†]These authors contributed equally.
[*]Corresponding author: Jia Meng, jia.meng@xjtlu.edu.cn.

Abstract

This document provides detailed data inclusion criteria, benchmark construction procedures, balance checks, train–test entity-overlap audits, model specifications, evaluation procedures, and interpretation protocols supporting the main manuscript.

Key words: MHC-I; tissue-specific preference; supplementary material; benchmark construction; data audit

The analyses below provide the full technical details supporting the condensed data description in Section 2 of the main manuscript.

S1. Detailed Data and Task Definition

S1.1. Defining a Tissue–MHC Prediction Context

One prediction context corresponds to one tissue and one MHC-I restriction. For example, lung with HLA-A*02:01 and liver with the same allele are different contexts because the tissue differs. Let t denote a tissue and m an MHC-I restriction. We define the set of observed contexts as

$$\mathcal{T} = \{\tau = (t, m)\}.$$

Each benchmark row is represented by

$$x_i = (p_i, \tau_i), \qquad y_i \in \{0, 1\},$$

where p_i is a nine-residue peptide and τ_i identifies one tissue–MHC context. A positive label, $y_i = 1$, indicates positive ligand evidence in the target context. A comparison label, $y_i = 0$, indicates positive evidence under the same MHC restriction and parent protein in at least one other tissue, but no report in the target context. Thus, the pseudo-negative label represents absence of recorded evidence rather than experimentally verified non-presentation.

Given (p, τ), a model parameterized by θ produces

$$s_\theta(p, \tau) \in \mathbb{R},$$

with larger values indicating stronger predicted evidence within tissue–MHC combination τ. Operationally, the model learns to rank a peptide reported in the target tissue above a comparison peptide matched by MHC restriction and parent protein. It is trained on individual peptide rows, while the paired construction controls example generation and evaluation. Because each tissue–MHC combination has its own prediction function, the benchmark evaluates previously represented combinations; it does not assess unseen tissues or unseen MHC restrictions.

S1.2. Pairing Peptides by Tissue, MHC Restriction, and Parent Protein

Each pair j contains one positive peptide p_j^+ and one pseudo-negative peptide p_j^- with evidence elsewhere but not in the target tissue. Let $c(p, m, u)$ be the number of tissues with positive source-record evidence for peptide p under MHC restriction m and parent protein u. The two rows share the same target tissue, MHC restriction, parent UniProt protein, and positive tissue-occurrence count:

$$t_j^+ = t_j^-, \qquad m_j^+ = m_j^-, \qquad u_j^+ = u_j^-.$$

and

$$c(p_j^+, m_j, u_j) = c(p_j^-, m_j, u_j).$$

This matching removes three easy explanations for a difference: the peptides cannot be separated merely because they use different MHC molecules, come from different proteins, or have different total numbers of positive tissue reports. The remaining difference can include tissue-associated processing and presentation, but also the identities of the reporting tissues, study design, detection sensitivity, and annotation completeness. The benchmark therefore measures tissue-specific preference in presentation, not a purified molecular mechanism.

We first calculate performance separately for every tissue–MHC combination and then give each combination equal weight, so common combinations do not dominate rare ones. We also report within-pair ordering accuracy:

$$\mathrm{Acc}_{\mathrm{pair}} = \frac{1}{N}\sum_{j=1}^{N} \mathbf{1}\Big[s_\theta(p_j^+, \tau_j) > s_\theta(p_j^-, \tau_j)\Big].$$

where N is the number of held-out pairs. This quantity is the fraction of pairs for which the peptide reported in the target tissue receives the higher score. AUROC measures ordering across all rows within a tissue–MHC combination, whereas within-pair accuracy directly evaluates the original MHC- and parent-matched comparison.

S1.3. Evidence Inclusion Criteria

The human and mouse benchmarks are derived from the full ligand export of the IEDB downloaded on July 4, 2026 (1). We apply the same biological inclusion rules to both species. We retain only clearly positive ligand records for MHC-I with an identifiable parent protein and a standard unmodified 9-amino-acid peptide:

- a qualitative measurement whose normalized text begins with *positive*;
- a MHC-I restriction, with both peptide source and host matching Homo sapiens for the human benchmark or Mus musculus for the mouse benchmark;
- a human restriction matching the four-digit HLA pattern `HLA-...*dd:dd` (the extractor also accepts the equivalent colon-free four-digit form) or a restriction from the mouse major histocompatibility complex (H-2; written as `H2` in the processed identifiers) beginning with `H2-`;
- a parent UniProt accession extracted from the molecule-parent web identifier (internationalized resource identifier; IRI) in the IEDB;
- a linear, unmodified peptide containing only the 20 standard amino acids;
- a peptide length of exactly nine residues; and
- deduplication before pairing by the exact peptide, restriction, tissue, and parent/source-protein annotation fields.

Source-tissue values are stripped of surrounding whitespace and retained as the exact names used in the IEDB; distinct nonempty tissue strings are not merged after extraction. Empty or missing-value-like fields required to define a tissue–MHC combination are removed before benchmark filtering. Pair-consistency checks require both rows to have the same nonempty tissue and restriction. Retained HLA and H-2 strings are used directly as identifiers after the pattern checks above. Records without a parent UniProt accession are excluded because parent-protein matching is a defining control of the benchmark. Restricting the study to unmodified canonical 9-residue peptides for MHC-I provides a consistent sequence space, but excludes variable-length ligands, post-translationally modified peptides, and presentation by MHC-II.

S1.4. Constructing MHC- and Parent-matched Peptide Pairs

For each target tissue–MHC combination, we compare a peptide reported in that tissue with another peptide from the same protein that is reported in a different tissue under the same MHC restriction. A fixed random state, 20260704, makes the following construction reproducible:

1. Group all eligible ligand records by MHC restriction m and parent UniProt protein u.
2. For each target combination (t, m, u), collect peptides with positive evidence in tissue t and calculate their positive tissue-occurrence counts.
3. From other tissues represented in the same (m, u) group, collect candidate comparison peptides with evidence elsewhere but not yet in the target tissue (pseudo-negative peptides), stratified by the same occurrence count.
4. Exclude every candidate with a positive report in the target tissue–MHC combination (t, m).
5. Within each target combination, parent protein, occurrence-count stratum, and label, use each peptide at most once. Set the number of pairs to the smaller of the eligible positive and pseudo-negative counts and sample both sets without replacement.
6. Randomly permute the selected pseudo-negatives and join them one-to-one with the selected positives.
7. Assign a persistent pair identifier, preserve the reported-tissue and protein annotations, and verify that every identifier has exactly one row per label, identical tissue, restriction, and parent UniProt values in both rows, and no target-tissue report for the pseudo-negative.

Every pair contributes one target-tissue peptide and one comparison peptide, so the dataset is deliberately balanced. This balance is useful for controlled ranking but does not resemble the natural frequency of presented peptides in a tissue sample. The reported scores therefore describe relative ranking within this constructed benchmark, not the probability that a peptide would be detected in a biological specimen.

The seven construction steps above define the matched-pair unit used throughout the occurrence-matched benchmark. Figure S1 then verifies the additional occurrence-count equality across every retained tissue.

S1.5. Human Benchmark

The occurrence-matched human benchmark contains 77 tissue–HLA combinations. For every matched pair, the positive and pseudo-negative peptides have the same number of positive tissue occurrences in the source records. This additional control prevents the model from separating the labels merely by learning that one member of a pair was reported in more tissues overall. Each retained combination contributes exactly 50 complete pairs to the fixed internal test set, and all other pairs form the training partition. Random state 20260704 makes this division reproducible.

Table S1 summarizes the Human task inventory and fixed split.

The 77 tissue–HLA combinations form an incomplete and long-tailed tissue-by-HLA coverage matrix. Performance is therefore calculated separately for each combination before averaging with equal weight; pooling all rows would overemphasize combinations with more examples and could conceal weak performance in sparsely represented contexts. The fixed internal test set excludes complete pair identifiers, but a peptide or parent protein may still occur in a different combination on both sides of the split. It therefore evaluates familiar tissue–HLA contexts rather than biological entities that are entirely absent from training.

Table S1. Size and task inventory of the occurrence-matched Human benchmark. Counts describe the deterministic saved split, not repeated fits: all 77 tissue–HLA tasks contribute 50 complete pairs to the fixed internal test, and the remaining pairs form the training file. This table defines the Human data scale used by all subsequent Human performance comparisons.

Property	Value
Tissue–HLA combinations	77
Tissues	13
HLA alleles	29
Minimum retained pairs per combination	100
Training pairs	21,489
Test pairs	3,850
Training rows	42,978
Test rows	7,700
Test pairs per combination	50

S1.6. Mouse Benchmark

The mouse benchmark uses the same tissue-specific preference task and occurrence-matching definitions: the two peptides share their H-2 restriction, parent protein, and total number of positive tissue occurrences. The supplied occurrence-matched files retain 11 tissue–H-2 combinations. Exactly 50 complete matched pairs per retained combination are assigned to the fixed internal test set using random state 20260704, and the remainder form the training partition.

Table S2 reports the corresponding Mouse task inventory and fixed split.

The benchmarks are intentionally parallel in construction but not equal in scale. After applying the same biological filters and occurrence-matching rule, the human files contain 77 retained tissue–HLA combinations, 13 tissue labels, 29 HLA alleles, and 25,339 pairs, whereas the mouse files contain 11 tissue–H-2 combinations, four tissues, four H-2 restrictions, and 2,639 pairs. The fixed test allocation is identical—50 pairs per retained combination—so the test-set difference of 3,850 versus 550 pairs follows exactly from the 77-versus-11 task inventories. The remaining difference reflects the number and coverage of eligible source records and retained tissue–MHC contexts, not different per-task sampling rules. Consequently, raw performance differences between species are descriptive and cannot be interpreted as species effects.

Table S2. Size and task inventory of the occurrence-matched Mouse benchmark. Counts describe the deterministic saved split, not repeated fits: all 11 tissue–H-2 tasks contribute 50 complete pairs to the fixed internal test, and the remaining pairs form the training file. This table defines the Mouse data scale used by subsequent mouse performance comparisons.

Property	Value
Tissue–H-2 combinations	11
Tissues	4
H-2 restrictions	4
Minimum retained pairs per combination	134
Training pairs	2,089
Test pairs	550
Training rows	4,178
Test rows	1,100
Test pairs per combination	50

S2. Detailed Data Balance and Leakage Audit

S2.1. Occurrence-balancing Audit

We audited the supplied training and test files pair by pair before model evaluation. A valid pair must contain exactly one row of each label; both rows must have the same tissue, MHC restriction, and parent UniProt identifier; and the two peptides must have the same *presentation_tissue_count*. All 27,978 pairs across the two species satisfy these conditions, with zero mismatched pairs in either partition. The complete Human benchmark contains 32,625 positive tissue occurrences in each label class, and the complete Mouse benchmark contains 2,807 in each label class. Equality also holds separately within every task in the training and test partitions (Table S3), including the complete frequency distribution rather than only its total. A score using only *presentation_tissue_count* has AUROC 0.5000 in every task (minimum, mean, and maximum all 0.5000). Thus, label identity cannot be inferred from total positive tissue-occurrence count in this benchmark.

Figure S1 resolves the complete-data occurrence totals by tissue. In all 13 Human tissues and all four Mouse tissues, the label-0 and label-1 totals coincide exactly; the maximum absolute tissue-level difference is zero.

Controlled shortcut versus residual sources of bias.

We use occurrence matching to remove one prespecified shortcut: a model cannot favor the label-1 peptide simply because it was reported positively in more tissues overall. This result should not be described as removal of all dataset bias. The construction does not balance the identities of the other reporting tissues, study or laboratory provenance, donor composition, tissue coverage, allele coverage, sample preparation, mass-spectrometry platform, detection depth, or annotation completeness. It also does not prevent an exact peptide or parent protein from recurring in another tissue–MHC combination. Statements about bias in the remainder of the paper therefore distinguish the eliminated occurrence-count shortcut from these residual observational and provenance effects.

S2.2. Entity Overlap across the Fixed Split

In both species, a held-out peptide is absent from the training rows of the tissue–MHC combination being evaluated, but it may occur in training rows for another combination. Table S4 makes this distinction explicit. For each test row, the audit first asks whether the same entity occurs in training for the current combination and, only if it does not, whether it occurs in another combination. The three columns are therefore mutually exclusive.

Table S3. Deterministic occurrence-balancing audit of the saved Human and Mouse training and fixed-test files. Occurrence totals sum *presentation_tissue_count* separately over the two label classes; a mismatch denotes a pair whose two rows have different occurrence counts. The table tests whether pair construction removed the occurrence-count shortcut and involves no model fitting, seeds, or performance aggregation.

Species	Partition	Pairs	Label 0/1 totals	Mismatches
Human	Training	21,489	27,706/27,706	0
Human	Test	3,850	4,919/4,919	0
Mouse	Training	2,089	2,204/2,204	0
Mouse	Test	550	603/603	0

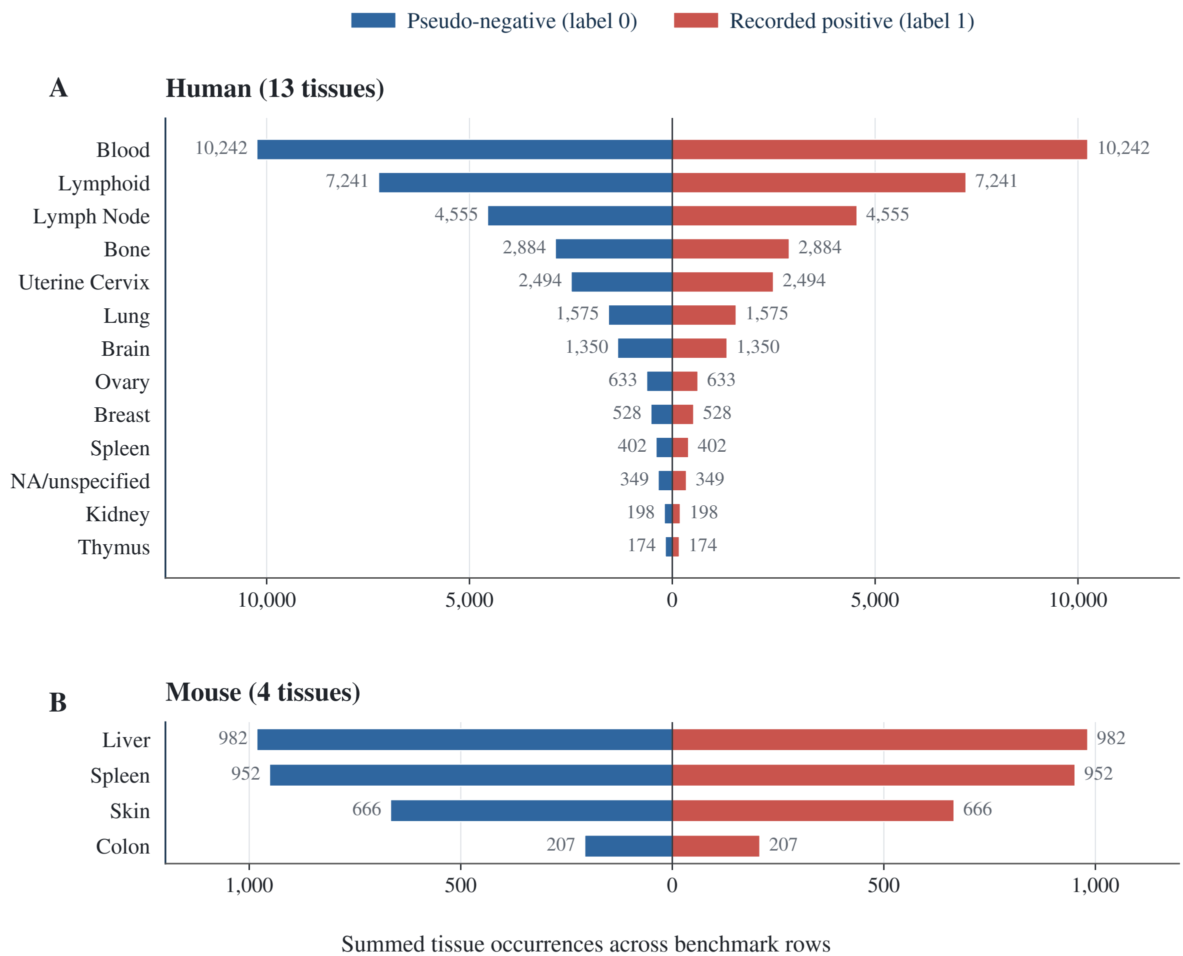

Fig. S1: Tissue-resolved occurrence-balancing audit on the complete occurrence-matched benchmark. **A**, Human totals across all 13 retained tissue labels. **B**, Mouse totals across all four retained tissues. For each tissue and label, the plotted quantity is the deterministic sum of *presentation_tissue_count* over all training and fixed-test rows; no model fitting or seed aggregation is involved. Blue bars extend left for pseudo-negatives (label 0), and vermilion bars extend right for recorded positives (label 1). Exact bilateral symmetry and the endpoint labels show that the two totals are identical within every tissue. Both panels use linear axes scaled to their species-specific ranges; cross-panel bar lengths should therefore not be compared directly.

Table S4 answers an entity-reuse question about the saved train/test split; Supplementary Table S5 instead defines the stages used for training-only model selection, final fitting, and fixed-test estimation. No separate cross-validation overlap table is reported because the tuning folds are not a generalization result and would duplicate the pair-grouping rule already stated in the evaluation protocol.

Thus, each test peptide is new to the specific tissue–MHC task being evaluated, although 65.35% of Human and 64.36% of Mouse test-row peptides occur in training for another task. Parent proteins recur more often, including within the current task. The fixed test is therefore an internal evaluation in familiar contexts, not an entity-disjoint or independent external validation.

S3. Detailed TissuePMHC Model

S3.1. Model Overview and Prediction of Tissue-Specific Preference

Peptide presentation contains sequence patterns at more than one biological level. Some features may recur across many tissues and HLA alleles, whereas others may reflect motifs associated with one allele. TissuePMHC therefore contains a global pathway that learns from all tissue–HLA combinations with additional tissue and allele supervision, and an HLA-specific pathway that learns only from combinations sharing one HLA allele. Each pathway produces a separate score for every tissue–HLA combination. Human and Mouse use separately tuned configurations selected by cross-validation confined to the corresponding training split. Both final models are fitted with the same three random initializations; unless an ensemble is explicitly identified, results summarize seed-level metrics rather than averaging row-level predictions.

Both pathways use a position-aware peptide encoder that preserves residue position and examines neighboring residues, followed by a separate output layer for each observed tissue–HLA combination. This structure allows combinations with many examples to support those with fewer examples without forcing all alleles to use an identical peptide representation. Because the final prediction depends on an output layer learned for a particular combination, the model is not designed for tissue–HLA combinations absent from training.

The remainder of this section gives the exact implementation needed for reproduction. Let h_i denote the numerical representation of peptide p_i, and let τ_i denote its tissue–HLA combination. Each pathway maintains a separate linear output for every available combination. The corresponding logit is

$$z_i = g_{\tau_i}(h_i),$$

and the associated presentation-preference probability is

$$q_i = \sigma(z_i),$$

where σ is the logistic sigmoid. During each mini-batch update, examples are routed to the output layer for their tissue–HLA combination, while encoder parameters are shared according to the pathway definition. The main objective is binary cross-entropy with logits, denoted BCE:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{B}\sum_{i=1}^{B}\left[y_i \log q_i + (1-y_i)\log(1-q_i)\right].$$

Although every benchmark pair contains one recorded-positive row and one pseudo-negative row, the model processes rows independently. Pair identifiers are not used during the forward pass or loss calculation.

S3.2. Peptide Encoding That Preserves Position and Examines Several Motif Lengths

The biological meaning of an amino acid depends on its position in a 9-mer peptide: anchor residues involved in MHC binding are not interchangeable with residues at other positions. The encoder therefore preserves all nine positions while examining local patterns of several lengths. For a peptide $p = (a_1, \ldots, a_9)$, each residue is mapped to a learned 16-dimensional embedding and combined with a learned positional embedding:

$$X = E(p) + P \in \mathbb{R}^{9\times d},$$

where $E(p)$ contains the residue embeddings and P is a trainable $9 \times d$ positional representation shared across peptides. These positional embeddings distinguish the same amino acid at different locations, which is important because anchor and supporting positions in MHC-I ligands are not interchangeable.

The model operates only on canonical 9-mer peptides. No parent-protein sequence, peptide-flanking sequence, explicit cleavage, transport, trimming, or loading features, or external binding score is supplied as input. Tissue and HLA information enter through the combination identifiers, combination-specific output layers, parameter-sharing structure, and auxiliary training labels.

To examine local sequence patterns spanning two, three, or five residues, we apply one-dimensional convolutions with widths

$$\mathcal{K} = \{2, 3, 5\}.$$

For each $k \in \mathcal{K}$, the input is transposed to a channels-first representation and processed as

$$H_k = \mathrm{ReLU}\left(\mathrm{Conv1D}_k(X)\right).$$

Each convolution produces 32 feature maps. Padding retains local context at the peptide ends. For even kernel widths, symmetric padding produces one additional position; the final position is therefore cropped to retain a length of nine. Each pathway consequently yields

$$H_k \in \mathbb{R}^{9\times 32}.$$

The three sets of local patterns are joined while preserving their sequence positions. We deliberately avoid reducing the peptide to a position-free maximum or average, because the same short motif

Table S4. Deterministic Human and Mouse entity-overlap audit for the occurrence-matched fixed internal test sets. The table asks whether held-out pair identifiers, peptide identities, or parent proteins reappear in current- or other-combination training rows; it is not a model-performance comparison. Counts are test-row occurrences, except for pair identifiers, which are counted once per pair. "Current combination" means the tissue–MHC combination of the test row, and the three outcome columns are mutually exclusive.

Species and held-out entity	Current-combination training	Other-combination training only	Absent from all training
Human pair identifier (3,850 pairs)	0 (0.00%)	0 (0.00%)	3,850 (100.00%)
Human peptide identity (7,700 rows)	0 (0.00%)	5,032 (65.35%)	2,668 (34.65%)
Human parent UniProt (7,700 rows)	1,112 (14.44%)	6,152 (79.90%)	436 (5.66%)
Mouse pair identifier (550 pairs)	0 (0.00%)	0 (0.00%)	550 (100.00%)
Mouse peptide identity (1,100 rows)	0 (0.00%)	708 (64.36%)	392 (35.64%)
Mouse parent UniProt (1,100 rows)	180 (16.36%)	726 (66.00%)	194 (17.64%)

can have a different meaning at a different position:

$$v = \text{Flatten}\,[H_2; H_3; H_5] \in \mathbb{R}^{9\cdot 32\cdot 3}.$$

The resulting vector is passed through a two-layer multilayer perceptron,

$$h = \text{MLP}(v) \in \mathbb{R}^{128},$$

Each hidden layer has 128 units with rectified linear unit activation. In the selected Human configuration, dropout with probability 0.35 is applied after the first hidden layer. Flattening the position-specific feature maps allows identical local patterns at different peptide positions to contribute differently to the representation. The multi-kernel encoder is tailored to peptides of this single length; convolution itself is not presented as a methodological novelty.

S3.3. Information Sharing Across All Combinations and Within HLA Alleles

The global pathway allows every tissue–HLA combination to contribute to a shared peptide representation. During training, auxiliary tissue and HLA classification objectives encourage this representation to retain biologically relevant grouping information. These objectives do not provide direct measurements of cleavage, transport, trimming, loading, or other antigen-presentation machinery. If $c_i^{(t)}$ and $c_i^{(m)}$ denote the tissue and HLA categories, respectively, the global objective combines BCE with categorical cross-entropy, denoted CE:

$$\mathcal{L}_{\text{global}} = \mathcal{L}_{\text{BCE}} + \lambda_t \mathcal{L}_{\text{CE}}\left(\widehat{c}^{(t)}, c^{(t)}\right) + \lambda_m \mathcal{L}_{\text{CE}}\left(\widehat{c}^{(m)}, c^{(m)}\right),$$

with training-CV-selected weights

$$\lambda_t = \lambda_m = 0.3.$$

The tissue and HLA predictions guide representation learning but do not enter the final peptide score. They provide auxiliary supervision rather than independent biological evidence.

The HLA-specific pathway shares information only among tissues with the same allele, preserving allele-associated peptide preferences that broad pooling could blur. For each HLA allele m, we define

$$\mathcal{T}_m = \{(t, m) \in \mathcal{T}\}.$$

A separate encoder is trained on rows belonging to $\mathcal{T}_m$, and each tissue–HLA combination in the group receives its own linear output layer. The objective for allele m is

$$\mathcal{L}_m = \frac{1}{|\mathcal{D}_m|} \sum_{i \in \mathcal{D}_m} \mathcal{L}_{\text{BCE}}\left(g_{\tau_i}^{(m)}(h_i^{(m)}), y_i\right).$$

This pathway has no auxiliary tissue or HLA classification objective. Because every example handled by one allele-restricted model shares the same allele, the pathway concentrates on allele-specific peptide patterns and variation among tissues under that restriction. It complements the global pathway: broad sharing can stabilize combinations with few examples, whereas allele-restricted sharing can preserve motif structure that unrestricted pooling may blur.

S3.4. Combining the Two Within-combination Orderings

The two pathways are trained separately, so equal raw scores need not have the same calibration in both. We therefore combine their rankings rather than their raw scores. Within each tissue–HLA combination, every candidate is converted to a percentile rank. For combination τ, let $q_{g,i}$ and $q_{h,i}$ be the two scores for row i:

$$r_{g,i} = \text{PercentileRank}_\tau(q_{g,i}),$$
$$r_{h,i} = \text{PercentileRank}_\tau(q_{h,i}).$$

The equal-weight rank-fused prediction for one training run is

$$s_i^{(b)} = \frac{r_{g,i} + r_{h,i}}{2}.$$

Both pathways receive equal weight; no tissue or test set is used to choose a preferred pathway. This procedure answers a ranking question, not an absolute probability question. A peptide's percentile depends on the other candidates scored in the same batch, so adding or removing candidates can change its final rank. Prospective use would therefore require either a predefined candidate set or a prespecified reference distribution.

S3.5. Independent Training Runs and Reporting Scope

Neural-network training can vary across random initializations. We therefore repeat the complete training procedure three times with seeds 20260704, 20260705, and 20260706. Within each run, the two pathway probabilities are converted to within-combination ranks and combined as defined above. The primary occurrence-matched results summarize the three seed-level task-macro metrics by their arithmetic mean and sample standard deviation; they do not average row-level predictions across seeds. A row-level prediction aggregate is included only when a table explicitly marks that result with a dagger. Because the main architecture tables already report run-to-run variation under this common protocol, no separate supplementary stability or averaging-gain table is retained.

S3.6. Species-specific Mouse Models

The mouse analysis tests whether the same prediction problem is learnable in a second species; no human parameters are transferred to mouse. Mouse implementations are trained independently on the 11 retained tissue–H-2 combinations and include a shared encoder, H-2 identifier and 34-residue pseudo-sequence conditioning, an identifier–pseudo-sequence hybrid, and matched dual-branch models with and without auxiliary tissue and H-2 supervision. The selected Mouse TissuePMHC configuration uses convolution widths 3 and 5, dropout 0.35, tissue and H-2 auxiliary-loss weights of 0.05 and 0.10, equal rank-fusion weights, a learning rate of 3×10^{-3}, and task-balanced sampling with 32 rows drawn per task in each balanced batch. It is trained for 25 epochs. The four H-2 pseudo-sequences are taken from the versioned NetMHCpan 4.1b MHC resource (2) and mapped from its `H-2-*` aliases to the benchmark's `H2-*` identifiers; the source-file hash is retained with the derived table. All mouse methods use the same fixed-split protocol and the same three seeds as the human experiments.

S4. Detailed Experimental and Evaluation Protocol

S4.1. Evaluation Questions

The occurrence-matched experiments ask whether peptide sequence retains tissue-associated signal after matching the MHC, source protein, and total positive tissue-occurrence count; how alternative model structures compare on the fixed Human benchmark; whether the same prediction problem is learnable in Mouse; how much signal is recoverable without tissue identity; and which residue–position patterns support the fitted Human model's decisions. The Human–Mouse analysis concerns replication of the benchmark definition, not transfer of one model between species.

Terminology layers.

We reserve *data partition* for the saved training and fixed-test files; *model selection* for pair-grouped folds formed only inside the training file; *fitting run* for one optimization run with one seed; *fixed-test evaluation* for scoring the locked model on the saved test rows; *prediction aggregation* for explicitly labeled row-level score combinations; and *statistical analysis* for task summaries, bootstrap intervals, and paired tests. Model architectures, optimization algorithms, split construction, and statistical procedures are therefore not described as interchangeable "evaluation modes."

S4.2. Fixed-Test Design and Biological Scope

The occurrence-matched experiments use one final evaluation protocol: a fixed internal train/test split with complete matched pairs kept intact. Hyperparameters are selected by three-fold, task-stratified, pair-grouped cross-validation confined to the training file; these folds are used only for model selection and are not reported as a generalization result. After each species-specific configuration is locked, three independent models are fitted on the complete training file and evaluated on the fixed test file. Table S5 separates model selection, final fitting, and fixed-test evaluation.

Figure S2 summarizes the single evaluation pathway used for all reported results.

What is actually predicted.

The benchmark is constructed from matched peptide pairs, but the main neural models do not receive two peptides simultaneously. They score one peptide row at a time using its sequence and tissue–MHC identity. Pair identifiers are used to create balanced recorded-positive and pseudo-negative examples and to keep the two rows of a pair in the same data partition. Training applies a separate present/absent loss to each row, and AUROC and AUPRC compare all positive and pseudo-negative rows within a tissue–MHC combination. Within-pair ordering accuracy asks the complementary question of whether the recorded-positive peptide receives the higher score. Keeping both members of a pair together protects data integrity; it does not turn the main training objective into a direct two-peptide ranking loss.

Saved internal test set.

For each retained tissue–MHC combination, the supplied split assigns exactly 50 complete pairs to the test file and all remaining pairs to the training file. Because a peptide is used at most once within a combination during pair construction, a test peptide is absent from training rows for that same combination. The sequence may nevertheless occur in another fitting combination and can then influence shared model parameters. This test set therefore measures internal discrimination in already represented biological contexts while allowing information sharing across combinations. Human and mouse use this same rule.

Independent training runs and reported uncertainty.

Every trainable human and mouse method is fitted three times with seeds 20260704, 20260705, and 20260706. All three fits use the same species-specific training file and are evaluated on the same fixed test file. Unless a table explicitly identifies a prediction ensemble, the reported central value is the arithmetic mean of the three seed-level metrics and the accompanying variation is their sample standard deviation. The seeds quantify optimization sensitivity only; they are not biological replicates and do not increase the number of evaluated tissue–MHC combinations.

Scope of the biological conclusion.

The fixed internal evaluation asks whether peptide sequence helps distinguish recorded positives from occurrence-matched pseudo-negatives in familiar tissue–MHC contexts. It does not evaluate a new tissue, MHC allele, tissue–MHC combination, study, or external cohort. Exact peptides and parent proteins may recur across different combinations on opposite sides of the fixed split. Training-only cross-validation selects hyperparameters but does not supply evidence for ordinary same-pool or peptide-separated generalization.

S4.3. Training and Baseline Models

All neural encoders and combination-specific output layers are optimized with AdamW. The selected Human TissuePMHC configuration is trained for 20 epochs with batch size 512, learning rate 2×10^{-3}, weight decay 10^{-4}, dropout 0.35, auxiliary tissue and HLA weights of 0.30, and gradient-norm clipping at 1.0. The selected Mouse configuration is trained for 25 epochs with task-balanced batches, learning rate 3×10^{-3}, weight decay 10^{-4}, dropout 0.35, and auxiliary tissue and H-2 weights of 0.05 and 0.10. The global and allele-specific pathways are trained independently for each random initialization. Test rows are used solely for final prediction and evaluation.

The Human and Mouse final models use random initializations 20260704, 20260705, and 20260706. Human tuning compares the default configuration, a learning-rate change, and a combined learning-rate/dropout/auxiliary-weight configuration by three-seed out-of-fold summaries. Mouse tuning evaluates staged architecture, sampling, loss-weight, fusion, optimization, and regularization choices. In both species the winning configuration is locked from training folds before the fixed test is evaluated once. Final training follows a fixed epoch schedule without test-based early stopping. Preprocessing and combination mappings are reconstructed from each fitting partition so that held-out fold labels do not influence training.

The reported comparison models test different scopes of information sharing across tissue–MHC combinations. A shared-encoder model learns one peptide representation with a separate output for each context; conditioned models add tissue and MHC identifiers or MHC pseudo-sequences; grouped and dual-branch models restrict part of the sharing by tissue or MHC. All reported trainable systems use the same species-specific tissue–MHC combinations, fixed split, and three seeds, except the row explicitly marked as a row-level prediction aggregate.

The conditioned baselines add tissue and MHC identifiers, an MHC pseudo-sequence representation, or both to a shared peptide encoder. Hard-sharing baselines restrict encoder sharing by tissue

Table S5. Evaluation protocol used for all occurrence-matched performance results. The rows separate the fixed data partition, training-only model selection, and three-seed final fitting; they are different layers rather than alternative evaluation modes. The table defines when test rows are accessible and what uncertainty the repeated fits quantify, but reports no model performance itself.

Reader-facing setting	Layer	What is done	Purpose and relationship to the other setting
Fixed internal test set	Data setting	For each retained tissue–MHC combination, 50 complete positive–pseudo-negative pairs are kept outside the training file. Every trained model is evaluated on this same fixed set.	Internal benchmark for familiar tissue–MHC contexts. It is not cross-validation and is not an external cohort.
Training-only hyperparameter selection	Model-selection setting	Three task-stratified, pair-grouped folds are formed only within the training file. Human configurations are ranked by task-macro out-of-fold AUROC with Worst-10 AUROC as the tie-breaker; Mouse configurations are ranked by the mean of task-macro AUROC and AUPRC.	Locks one species-specific TissuePMHC configuration before the fixed test is opened. These out-of-fold scores are not used as evidence of strict peptide, tissue, allele, study, or cohort generalization.
Three independent training runs	Model-estimation procedure	Each trainable method is fitted three times on the same training file with seeds 20260704, 20260705, and 20260706, then evaluated on the same fixed test file. Reported stability summaries use the arithmetic mean and sample standard deviation of seed-level metrics.	Quantifies sensitivity to initialization. The three runs are technical repetitions, not independent biological samples and not three test splits.

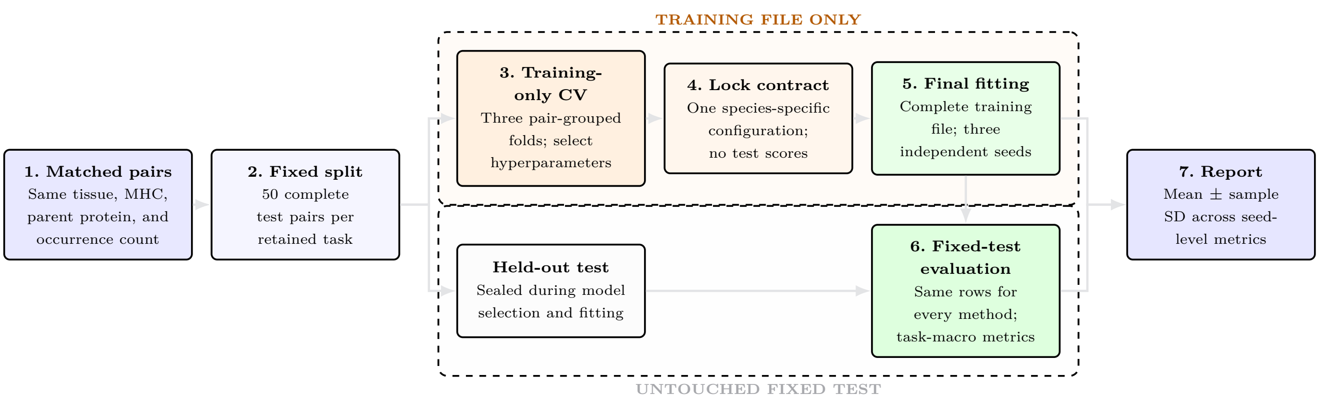

Fig. S2: Evaluation workflow for the occurrence-matched experiments. The fixed split creates two visually separated information lanes. The upper lane uses only the training file for three-fold, pair-grouped hyperparameter selection, contract locking, and three independent final fits. The lower lane keeps the fixed test sealed until the selected model is fitted. Human and Mouse use the same 50-pair-per-task test allocation and final-training seeds. Cross-validation is therefore a model-selection operation, not a reported ordinary-cross-validation result; every performance result in the main tables and figures comes from the untouched fixed test.

or by MHC group. Selective-grouping and dual-branch systems combine a global pathway with a grouped pathway; fixed-average variants weight the two pathways equally, whereas the validation-delta and validation-softmax variants derive their branch weights from the fitting-stage validation summaries. The two-view MLP is the dual-branch model with flattened residue embeddings, and the multi-kernel TissuePMHC model replaces that peptide encoder with the position-aware convolutional encoder defined above.

The formal component study starts from the locked full TissuePMHC model and removes one element at a time. The no-MHC-branch condition retains only the fitted global pathway. The no-rank-fusion condition retains both fitted pathways but averages their raw probabilities instead of their within-task percentile ranks. The no-auxiliary condition removes the tissue and MHC auxiliary losses while retaining both pathways and rank fusion. The no-multikernel condition replaces the convolutional encoder with a position-preserving flattened multilayer perceptron while retaining the remaining training and fusion contract. All four conditions use the same fixed split, task inventory, final-training seeds, optimizer, and species-specific epoch budget as the full model.

The trainable tissue-blind control retains the multi-kernel peptide encoder and one output head per MHC restriction but receives no tissue identifier and no tissue–MHC task identifier. It uses the same species-specific training rows, optimizer, epoch count, and final-training seeds. Mouse uses row-wise minibatches for this control so that a tissue-task-balanced sampler does not reintroduce tissue identity through the sampling process. Identical peptide–MHC queries are required to receive equal scores across tissues up to a prespecified numerical tolerance of 10^{-6}.

The reported multi-task comparisons include multi-gate mixture-of-experts (MMoE) (3) and DB-MTL (4). The dual-branch neural comparisons use flattened residue embeddings or the multi-kernel encoder and vary the sharing scope, auxiliary supervision, and validation-derived or fixed fusion weights. These definitions describe only systems retained in the reported comparison table.

S4.4. Performance Metrics and Statistical Analysis

Unless stated otherwise, performance estimates (including means, differences, interval endpoints, and sample standard deviations) are reported to four decimal places. Empirical percentages are reported to two decimal places, whereas conventional nominal levels such as 95% retain their standard whole-percentage form. Counts are integers with thousands separators. Exact design constants, identifiers, and hyperparameters retain the precision needed to specify them, and figure-axis tick labels may use shorter values for readability.

Comparability contract.
Data-audit tables contain deterministic counts and do not have training repeats. Model rows are comparable within species because they use one fixed test, one task inventory, and the same three seeds; a dagger identifies the only row-level prediction aggregate. Task, tissue, and allele summaries first average a task over the three seeds and only then aggregate tasks with equal weight. The combined main-text table aligns shared model families across species but does not treat their numerical difference as a controlled species effect.

The main question is whether peptides reported in the target tissue rank above comparison peptides with the same MHC restriction and parent protein. AUROC is calculated separately for every tissue–MHC combination and then averaged with equal weight, preventing data-rich tissues or alleles from dominating:

$$\mathrm{MacroAUROC} = \frac{1}{|\mathcal{T}_{\mathrm{valid}}|} \sum_{\tau \in \mathcal{T}_{\mathrm{valid}}} \mathrm{AUROC}_{\tau},$$

where $\mathcal{T}_{\mathrm{valid}}$ contains the prespecified combinations with both labels available. AUPRC is averaged in the same manner:

$$\mathrm{MacroAUPRC} = \frac{1}{|\mathcal{T}_{\mathrm{valid}}|} \sum_{\tau \in \mathcal{T}_{\mathrm{valid}}} \mathrm{AUPRC}_{\tau}\,.$$

Because the paired dataset has a constructed 1:1 label ratio, AUPRC is interpreted relative to the benchmark prevalence within each combination and is not compared directly with AUPRC from naturally imbalanced clinical cohorts.

Within-pair ordering accuracy is the fraction of matched pairs for which the recorded-positive peptide receives the higher score. Accuracy, F1, and MCC are secondary because they require a single decision threshold and the final score is not a calibrated biological probability.

For each random initialization, metrics are calculated separately for every tissue–MHC combination on the fixed test file and then averaged with equal task weight. The three seed-level macro metrics are summarized by their arithmetic mean and sample standard deviation. A row-level prediction ensemble is reported only when explicitly named; otherwise "three-run mean" refers to a mean of metrics rather than averaged predictions.

For the task-wise tuned-versus-untuned comparison in Supplementary Figure S5, uncertainty in the mean AUROC change is obtained by resampling complete tasks 10,000 times. Two-sided paired Wilcoxon signed-rank tests are calculated separately for Human and Mouse and adjusted together by Holm's method. Repeated training seeds quantify optimization sensitivity and are not treated as independent biological samples. Because tasks may share a tissue, allele, proteins, or related peptides, these summaries are descriptive rather than patient-cohort inference.

For the formal component ablations, each task metric is first averaged across the three seeds and then differenced against the full model on the same task. Mean-difference intervals use 10,000 bootstrap resamples of complete tasks. Two-sided paired Wilcoxon tests are Holm-adjusted over the four component contrasts separately within Human and Mouse. The trainable MHC-only control is a separate prespecified contrast and is reported with its exact paired Wilcoxon test and task-bootstrap interval.

S4.5. Post-hoc Expected Integrated Gradients Protocol

Sequence attribution is performed only for the locked human TissuePMHC configuration, after model selection and fixed-test performance evaluation. The analysis loads the fitted weights for the two differentiable E29 pathways—the globally shared pathway with auxiliary supervision and the HLA-restricted pathway—under each of the same three final-training seeds. For a peptide row x in task τ, the target is the corresponding pre-sigmoid branch logit $f_\tau(x)$. Residue identities are represented as differentiable one-hot inputs and passed through the fitted embedding projection. Expected Integrated Gradients averages integrated gradients over an empirical, task-conditional background B_τ:

$$\phi_i(x) = \mathbb{E}_{x' \sim B_\tau}\left[(x_i - x'_i)\int_0^1 \frac{\partial f_\tau(x' + \alpha(x - x'))}{\partial x_i}\, d\alpha\right].$$

This task-conditional Expected Integrated Gradients procedure is an empirical-baseline attribution approximation for differentiable networks rather than an exact enumeration of Shapley coalitions (5; 6; 7).

For each of the 77 Human tissue–HLA tasks, the background contains 10 complete training pairs and the explained subset contains 20 complete fixed-test pairs, both selected reproducibly with analysis seed 20260805. Positive and pseudo-negative members remain paired. The integral is evaluated deterministically by the trapezoidal rule with 128 steps. The design therefore explains 1,540 unique test pairs, or 3,080 rows per fitted branch and seed; across two branches and three seeds it yields 18,480 row-level branch explanations. Per-residue attributions sum over the one-hot channel and remain on the logit scale.

Quality control has three parts. First, predictions reconstructed from the loaded checkpoints must reproduce the saved per-seed fixed-test predictions. Second, attribution completeness is audited by comparing each branch logit with its empirical-background expected logit plus the summed attributions. Third, residue–position maps are compared across training seeds by Spearman correlation. Pair-matched positional effects are computed from the attribution assigned to the observed residue in each recorded-positive peptide minus that of its matched pseudo-negative, first within branch, seed, and task and then by equal-weight descriptive averaging. HLA-stratified maps analogously show the recorded-positive minus pseudo-negative mean attribution for each residue–position cell.

The final TissuePMHC score averages within-task percentile ranks from the two branches. Because percentile ranking is non-differentiable and cohort-dependent, no exact attribution decomposition of that fused score is claimed. The branch consensus is a descriptive arithmetic mean of the two audited branch summaries. The explained fixed-test subset is used only for post-hoc interpretation and does not alter fitting, hyperparameter selection, or reported predictive performance. These Expected-IG attributions describe model dependence conditional on the chosen task-specific background; they do not identify causal tissue biology.

S4.6. Single-residue In Silico Mutagenesis Validation

The Expected-IG patterns are checked by exhaustive single-residue ISM on exactly the same 3,080 human fixed-test rows, 1,540 complete pairs, 77 tasks, and three fitted seeds. For every 9-mer x, each position i is replaced in turn by each of the other 19 canonical amino acids. For branch $b \in \{g, h\}$, representing the global auxiliary and HLA-specific pathways, the logit perturbation is

$$\Delta_{i,a}^{(b)}(x) = f_\tau^{(b)}(x_{i\to a}) - f_\tau^{(b)}(x).$$

This gives 171 mutants per peptide and 526,680 distinct peptide–position–substitution rows; scoring both branches under all three seeds requires 3,160,080 mutant branch evaluations. No model is refitted for this analysis.

To evaluate the non-differentiable final output, each mutant replaces its original row in the complete fixed-test reference distribution for task τ. Its global and HLA-specific probabilities are independently converted to within-task percentile ranks by the same average-tie rule used by TissuePMHC, and the two ranks are averaged. The final perturbation $\Delta_{i,a}^{(F)}$ is this replacement rank-fusion score minus the saved original score. Position sensitivity is the mean $|\Delta|$ over the 19 alternatives and represented peptides. Classification-support loss is defined as $-(2y-1)\Delta$, so a positive value means that mutation moves a recorded-positive or pseudo-negative row against its assigned label.

The prespecified canonical-anchor comparison averages P2 and P9 sensitivity within each peptide and subtracts the mean over the other seven positions. A one-sided paired Wilcoxon test evaluates whether this difference is positive; 2,000 peptide bootstrap resamples provide a 95% interval, and Benjamini–Hochberg correction covers the three output scales and three label strata. Expected-IG–ISM agreement is measured for each branch by Spearman correlation between the observed-residue Expected-IG attribution and the mean ISM mutation loss, $-19^{-1}\sum_a \Delta_{i,a}^{(b)}$, over sample–position observations. Seed stability is separately computed over the 3,420 position–original-residue–mutant-residue mean effects. ISM still describes the response of the fitted model, and a computational substitution is not evidence of an in vivo causal effect.

To test tissue-task dependence while holding both sequence and HLA fixed, we perform a counterfactual same-HLA comparison. For each of the 28 Human HLA alleles represented in at least two tissues (76 tissue–HLA tasks), the explained test rows are deduplicated by peptide sequence within HLA, giving 2,939 HLA-counted unique peptides. Every peptide and each of its 171 single-residue substitutions are then scored under every represented tissue task for that HLA. Original and mutant probabilities are inserted separately into the corresponding fixed-test task distribution, and the global- and HLA-branch percentile-rank shifts are averaged. This produces 1,786,779 context–mutation rows per three-seed ensemble and 5,360,337 seed-specific rows. For mutation unit j, tissue dispersion and seed dispersion are

$$D_{\mathrm{tissue},j} = \operatorname*{mean}_{\tau<\tau'} \left|\bar{\Delta}_{\tau j} - \bar{\Delta}_{\tau' j}\right|,$$

$$D_{\mathrm{seed},j} = \operatorname*{mean}_{\tau} \operatorname*{mean}_{s<s'} \left|\Delta_{\tau j s} - \Delta_{\tau j s'}\right|,$$

where $\bar{\Delta}$ averages the three seeds and the comparisons are restricted to tissues sharing one HLA. Values are first averaged over substitutions and positions within peptide. A one-sided paired Wilcoxon test asks whether tissue dispersion exceeds seed dispersion across peptides for each HLA; 2,000 peptide bootstrap resamples give an interval for the mean difference, and Benjamini–Hochberg correction covers the 28 HLA tests. Tissue-pair Spearman correlations summarize shared mutation ordering, while correlations of tissue-contrast vectors across seeds assess reproducibility.

S4.7. Cross-species Interpretation and Reproducibility

The Human and Mouse benchmarks use the same occurrence-matched pair definition, the same 50-pair-per-task fixed-test allocation, and the same three training seeds. They differ in the number of retained tissue–MHC combinations, MHC system, and source-data composition. Human contains 77 tasks and 25,339 pairs, whereas Mouse contains 11 tasks and 2,639 pairs. The sevenfold test-size difference is therefore determined by the sevenfold difference in retained tasks, not by different test quotas. The larger difference in training size reflects the eligible pair inventory remaining after the shared biological and occurrence-matching rules. Concordant findings support replication of benchmark learnability but do not establish a species-independent molecular mechanism or a controlled species comparison.

Both species use random initializations 20260704–20260706. Cross-species comparisons emphasize above-random signal and the value of structured information sharing. Numerical differences are not interpreted as species effects because data scale, tissue and allele coverage, and model selection differ.

For reproducibility, each experiment records the fixed training and test files, training initialization, model configuration, and command-line arguments. Reports include epoch count, batch size, optimizer, learning rate, weight decay, dropout, training time, hardware, and software versions. Predictions are saved at row level with tissue–MHC combination, label, pair identifier, seed, and score.

Seed-specific settings and elapsed times are retained with the result files, and the three-seed aggregation records the seeds and the sample-standard-deviation convention. Training-only fold assignments, out-of-fold predictions, configuration rankings, and the locked contracts are also retained. These tuning folds are not presented as a strict-generalization benchmark.

S4.8. Data and Code Availability

The source code, data-processing, model-training, evaluation, and figure-generation scripts, together with the compact datasets distributed with this study and instructions for obtaining the omitted large IEDB export, are publicly available at `https://github.com/AmamiyaHitomi/tissuePMHC`. The human interpretation materials additionally document the explained pair identities, empirical-background design, per-seed and per-branch residue attributions, exhaustive ISM substitution effects, replacement rank-fusion scores, Expected-IG–ISM concordance, anchor-position tests, prediction-reproduction checks, seed-stability summaries, and plotting inputs. Raw material derived from the IEDB and third-party predictor assets remains subject to original licensing terms; NetMHCpan executables are not redistributed.

S5. Supplementary Results

S5.1. HLA-stratified Attribution Patterns

The paired positional analysis in the main manuscript identifies P2, P3, and P9 as the most influential positions. Supplementary Figure S3 further shows that this positional importance is not explained by one pooled residue pattern. For HLA-A*02:01, L, P, and Y at P2 and V, L, and Y at P9 receive positive contrasts, whereas HLA-A*24:02 and HLA-B*07:02 exhibit different residue–position structures. Each panel still combines tissues, source studies, proteins, and observational sampling; the maps therefore describe allele-associated dependence of the fitted model rather than a causal biochemical mechanism.

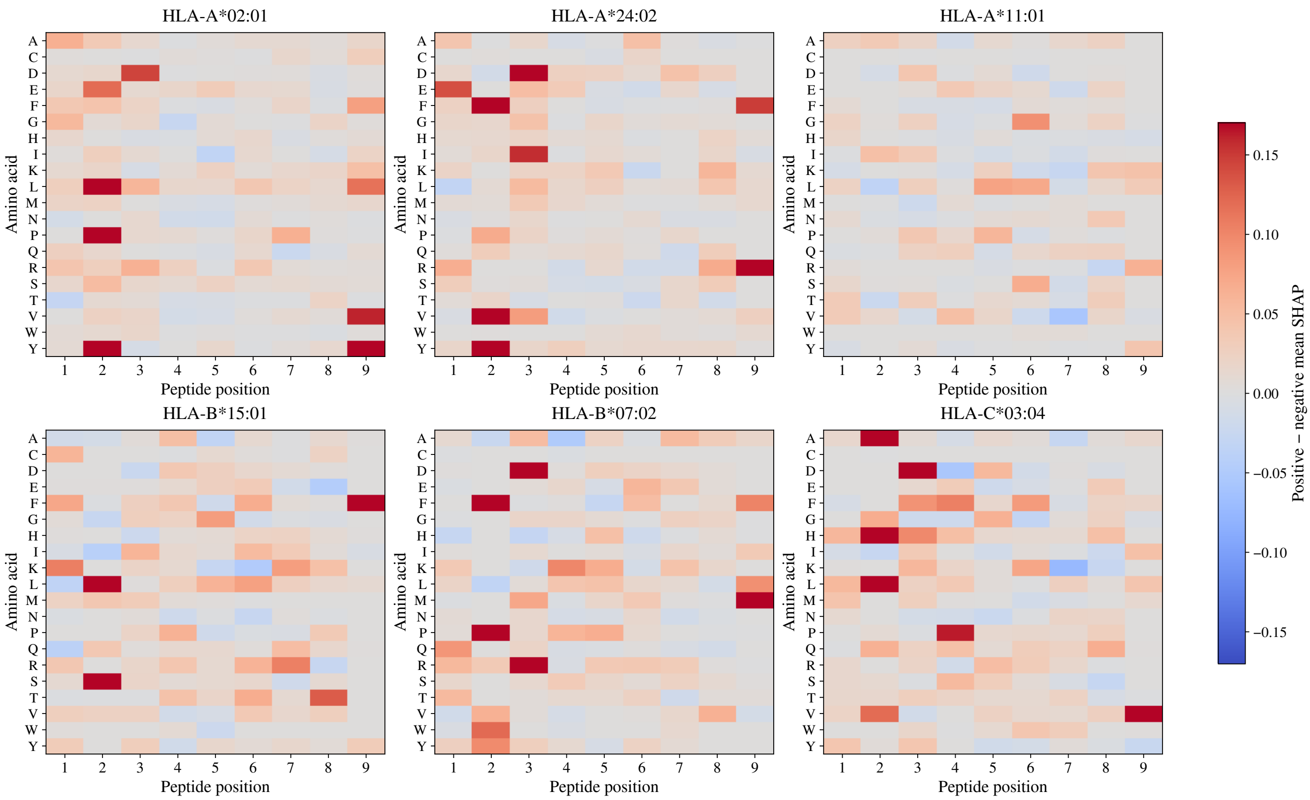

Fig. S3: HLA-stratified residue–position attribution maps for six well-represented Human alleles. Each cell is the mean branch-consensus attribution among recorded-positive rows minus that among occurrence-matched pseudo-negative rows for one amino acid and peptide position, aggregated across the three final-training seeds and the represented tissue–HLA tasks. Red indicates that the fitted branches associate that residue–position cell more strongly with the recorded-positive label; blue indicates the reverse. Blank-looking neutral cells are near zero, not missing measurements. Shared color limits permit visual comparison across panels. The maps describe model dependence conditional on task-specific empirical backgrounds and do not establish causal binding or antigen-processing effects.

S5.2. Counterfactual Same-HLA Tissue Analysis

The same-HLA counterfactual analysis fixes peptide, HLA, position, and substitution while changing the represented tissue task. Tissue dispersion exceeds training-seed dispersion for all 28 multi-tissue human HLA alleles, with peptide-bootstrap intervals above zero. P2, P3, and P9 have the largest pooled tissue dispersion (0.3074, 0.1920, and 0.1798). Supplementary Figure S4 shows both the allele-level dispersion ratios and a representative HLA-A*02:01 tissue perturbation fingerprint. These results support fitted-model tissue conditioning but do not identify a causal antigen-processing mechanism.

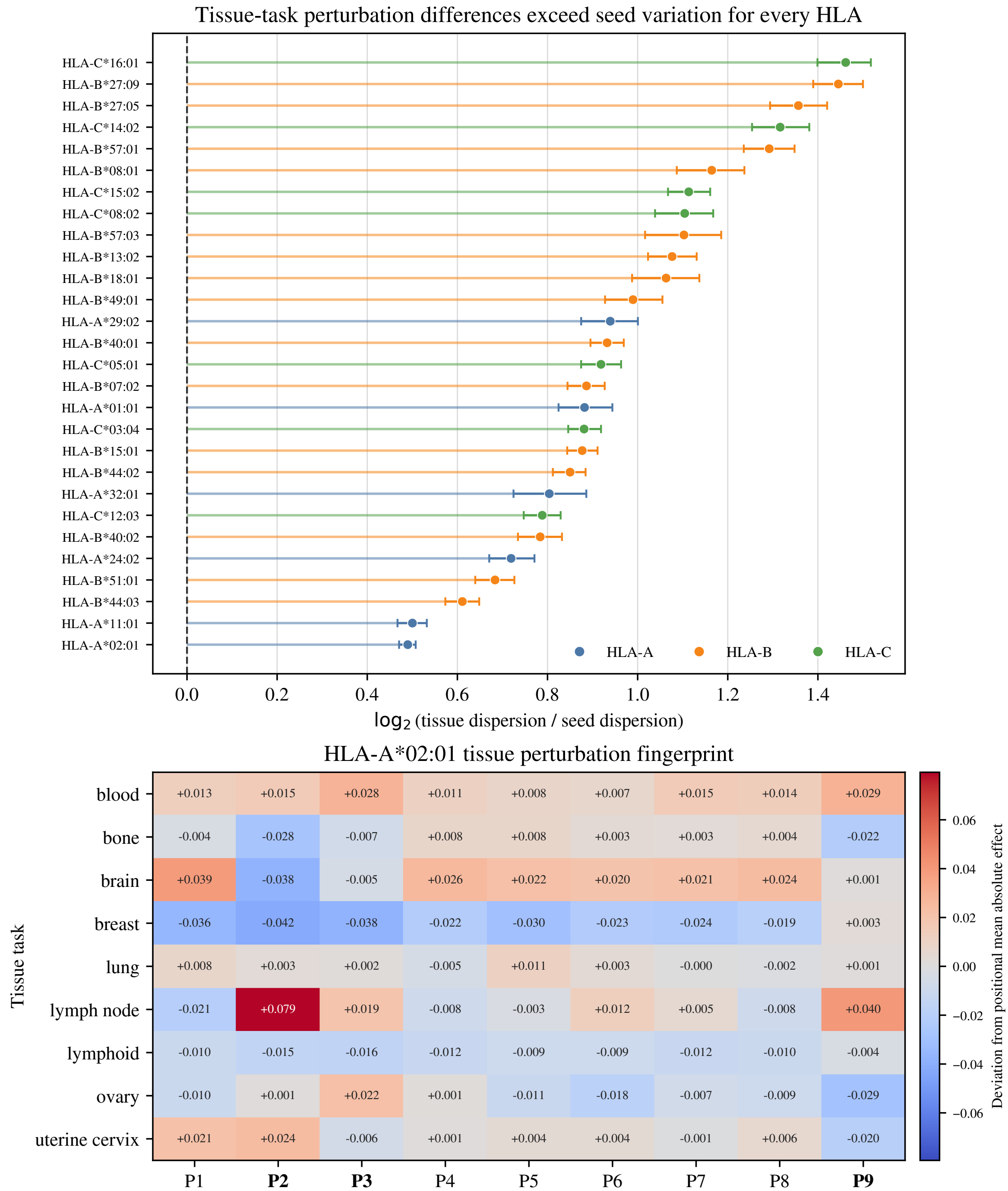

Fig. S4: Counterfactual same-HLA comparison of single-residue perturbation effects across Human tissue tasks. Each fixed peptide, position and substitution is scored under every represented tissue task for the same HLA. (a) HLA-level $\log_2$(tissue/seed) dispersion ratios for the final rank-fusion perturbation effect: dots are point estimates, horizontal bars are peptide-bootstrap 95% intervals and colour denotes HLA locus; the dashed line marks equal tissue and seed dispersion. (b) A representative HLA-A*02:01 tissue perturbation fingerprint. Each cell is the three-seed mean absolute rank-fusion effect at one peptide position in one annotated tissue task, centred by the allele's across-tissue mean for that position. Red and blue denote above- and below-baseline sensitivity; P2, P3 and P9 are emphasized. These contrasts establish fitted-model task dependence under fixed peptide and HLA inputs; they do not identify a causal tissue-processing mechanism.

S5.3. Detailed Tissue and Allele Heterogeneity

Performance varies across the 77 Human tasks. The lowest individual task is brain–HLA-A*11:01 (AUROC 0.6077), whereas the highest is breast–HLA-A*02:01 (0.9640). Supplementary Table S6 reports the corresponding tissue-level summaries. These values are descriptive because tissues differ in HLA inventory, source studies, peptide composition, and training coverage.

The 11 Mouse tissue–H-2 tasks span AUROC 0.5016–0.9180. Supplementary Table S7 reports all task values and the corresponding H-2-restriction means. Each restriction is represented in a different tissue subset, so the table does not estimate controlled H-2 effects.

Human performance is also heterogeneous across HLA loci and four-digit typings. Supplementary Table S8 reports the five lowest and five highest allele means. Unequal tissue and training inventories prevent interpretation as a controlled allele ranking.

Table S8. Lowest and highest Human four-digit HLA typings by mean fixed-test AUROC of tuned TissuePMHC. Each task is first averaged over the same three seeds on the occurrence-matched fixed test; allele means then weight the represented tissue–HLA tasks equally. Typings are selected solely by AUROC, five from each tail, while AUPRC is shown as a secondary metric. Task counts expose unequal tissue coverage, so the table is a descriptive extreme-value summary rather than a controlled allele ranking.

Group	HLA typing	Tasks	AUROC	AUPRC
Lowest	HLA-A*29:02	2	0.6500	0.6404
Lowest	HLA-A*11:01	3	0.6550	0.6468
Lowest	HLA-B*44:03	3	0.7036	0.6999
Lowest	HLA-B*49:01	2	0.7296	0.7105
Lowest	HLA-B*40:02	3	0.7442	0.7178
Highest	HLA-C*16:01	2	0.9351	0.9312
Highest	HLA-B*27:09	2	0.9143	0.9176
Highest	HLA-B*27:05	2	0.9120	0.9082
Highest	HLA-C*14:02	2	0.9119	0.9177
Highest	HLA-C*05:01	3	0.9010	0.8968

Training-only tuning increases mean task AUROC by 0.0054 in Human and 0.0489 in Mouse. Supplementary Figure S5 reports the complete task-wise distributions, bootstrap intervals, and paired tests. These statistics describe the fixed task inventories rather than independent biological samples.

S5.4. Detailed Tissue-independent Neural Control

Removing tissue identity while retaining peptide and MHC information creates an intentionally ambiguous prediction problem: the same peptide–MHC query may be positive in one tissue and a pseudo-negative in another. Supplementary Table S9 quantifies this label structure. Among queries represented by multiple tissue-specific rows, 95.18–99.04% contain both labels across the species and partition summaries.

Table S9. Deterministic label-conflict audit for controls without tissue information. Queries are defined by peptide sequence and MHC restriction. "One" and "multiple" count queries with one or multiple retained labels; conflicting queries contain both labels. Conflict percentages use multiple-label queries as the denominator.

Partition	Rows	Unique queries	One	Multiple	Conflict	%
Human benchmark						
Training pool	42,978	27,063	12,572	14,491	14,168	97.77
Fixed test	7,700	7,012	6,327	685	652	95.18
Complete benchmark	50,678	29,662	10,441	19,221	18,882	98.24
Mouse benchmark						
Training pool	4,178	2,934	1,735	1,199	1,175	98.00
Fixed test	1,100	996	892	104	103	99.04
Complete benchmark	5,278	3,338	1,487	1,851	1,833	99.03

Supplementary Table S10 reports the fitted MHC-only convolutional control. It loses 0.5211 mean task AUROC in Human and 0.6026 in Mouse and is worse than full TissuePMHC in every evaluated task. The below-random values reflect the cross-tissue label ambiguity rather than ordinary reverse generalization. Input-invariance checks confirm that identical peptide–MHC queries receive equal scores across tissues up to the prespecified numerical tolerance.

Table S10. Trainable MHC-only control on the occurrence-matched fixed tests. AUROC and AUPRC are task-macro mean $\pm$ sample standard deviation across seeds 20260704–20260706. Δ AUROC is candidate minus full TissuePMHC after averaging each task across seeds; brackets give a 10,000-resample task-bootstrap 95% confidence interval. W/T/L counts task-wise positive/zero/negative deltas. The exact two-sided paired Wilcoxon test is the prespecified MHC-only contrast. Identical peptide–MHC queries are constrained only by model inputs, not by post-hoc score averaging.

Model	AUROC	AUPRC
Human benchmark		
Full TissuePMHC	0.8136 ± 0.0021	0.8126 ± 0.0021
Trainable MHC-only CNN	0.2925 ± 0.0062	0.3932 ± 0.0040
Δ AUROC $= -0.5211$ [95% CI $-0.5489, -0.4927$]; W/T/L = 0/0/77; $p = 2.46 \times 10^{-14}$.		
Mouse benchmark		
Full TissuePMHC	0.8194 ± 0.0157	0.8279 ± 0.0121
Trainable MHC-only CNN	0.2168 ± 0.0052	0.3686 ± 0.0064
Δ AUROC $= -0.6026$ [95% CI $-0.7183, -0.4617$]; W/T/L = 0/0/11; $p = 0.0010$.		

Table S6. Human tissue-level fixed-test performance of tuned TissuePMHC on the occurrence-matched data. The table describes heterogeneity rather than a controlled biological tissue ranking. Each of the 77 tasks is first averaged over the same three seeds; tissue means then weight the available HLA tasks equally. Task counts expose the unequal HLA inventory across tissues. Tissues are ordered by AUROC from top to bottom in the left block and then the right block; the blank source label is reported as "NA/unspecified."

Tissue	Tasks	AUROC	AUPRC	Tissue	Tasks	AUROC	AUPRC
Brain	4	0.7314	0.7393	Blood	26	0.8265	0.8345
NA/unspecified	1	0.7384	0.7261	Uterine cervix	5	0.8328	0.8262
Thymus	1	0.7407	0.7548	Bone	1	0.8627	0.8762
Lung	7	0.7476	0.7417	Lymph node	5	0.8737	0.8583
Spleen	1	0.7765	0.7834	Kidney	1	0.8769	0.9003
Lymphoid	23	0.8129	0.8032	Breast	1	0.9640	0.9657
Ovary	1	0.8129	0.8200				

Table S7. Mouse heterogeneity of tuned TissuePMHC on the occurrence-matched fixed test. **A**, All 11 tissue–H-2 task values, each averaged over the same three seeds and ordered by AUROC. **B**, Descriptive H-2-restriction means formed after the seed-level task averages, with tasks weighted equally. Task counts expose the available tissue inventory. The table describes internal heterogeneity and cannot isolate tissue or H-2 effects because the represented contexts differ.

A. Tissue–H-2 tasks

Tissue	H-2	AUROC	AUPRC
Colon	H2-Db	0.5016	0.5025
Liver	H2-Db	0.6754	0.6695
Skin	H2-Kb	0.7821	0.8124
Liver	H2-Kb	0.7999	0.8186
Spleen	H2-Kb	0.8461	0.8524
Spleen	H2-Kd	0.8693	0.8676
Skin	H2-Kk	0.8760	0.8920
Liver	H2-Kk	0.9113	0.9126
Spleen	H2-Kk	0.9161	0.9220
Liver	H2-Kd	0.9180	0.9137
Skin	H2-Kd	0.9180	0.9440

B. H-2 restriction groups

H-2 restriction	Tasks	AUROC	AUPRC
H2-Db	2	0.5885	0.5860
H2-Kb	3	0.8094	0.8278
H2-Kd	3	0.9018	0.9085
H2-Kk	3	0.9011	0.9089

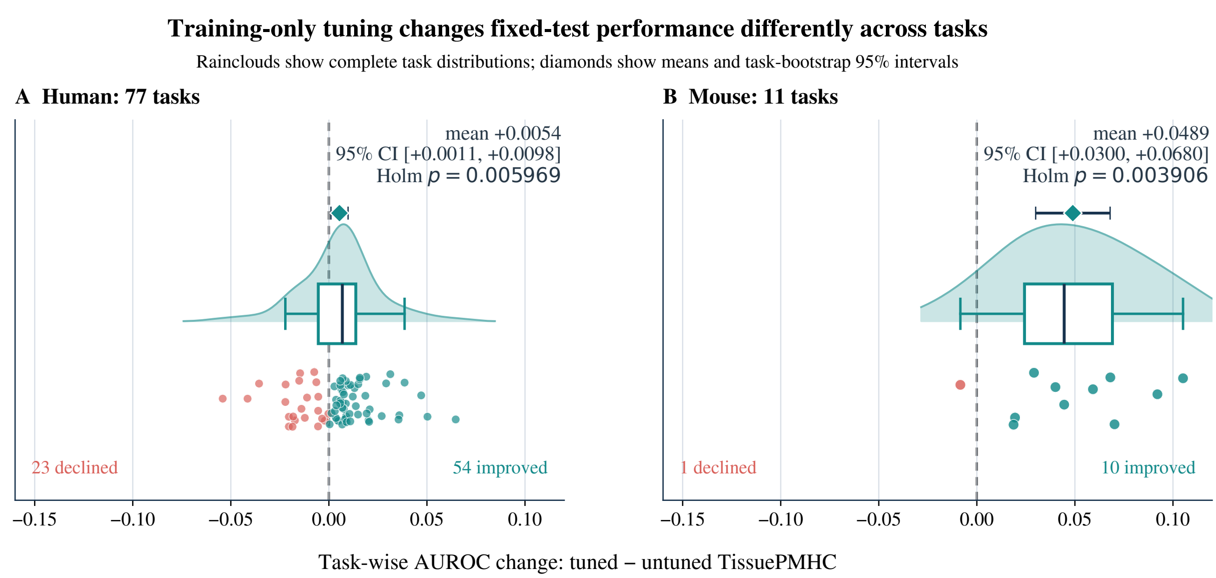

Fig. S5: Task-wise change in fixed-test AUROC after training-only hyperparameter tuning. Each point is the difference between tuned and untuned TissuePMHC after first averaging the same task over seeds; positive values favor the tuned configuration. Open diamonds show the mean change, and horizontal bars show 95% intervals from 10,000 bootstrap resamples of complete tasks. Two-sided paired Wilcoxon signed-rank tests were performed separately for Human and Mouse, then adjusted together by Holm's method (Human: 77 tasks, mean +0.0054, 54 improved, adjusted $p = 0.0060$; Mouse: 11 tasks, mean +0.0489, 10 improved, adjusted $p = 0.0039$). The intervals and tests describe this internal fixed-test task inventory and are not biological-replicate or cohort-level inference.

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